Refraction Er

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Refraction = #2 cause of visual impairment in India!
Super important topic!

Refraction Errors | Ophthalmology | 3rd Year | MBBS

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Definitions, Types + Optics Basics

Emmetropia = normal eye.
Ametropia = refractive error. Learn the types!

DEFINITION

"Refractive error = optical state where parallel rays of light fail to focus on the retina with accommodation at rest." -- Khurana

Optics Basics -- The Eye as a Lens System

Component	Details
Total refractive power	+58 to +60 D (total eye)
Cornea	+43 D (2/3 of total power -- most important!)
Lens	+15 to +20 D (remaining 1/3, adjustable)
Axial length (normal)	24 mm (1 mm change = 3 D change!)
Far point (emmetropia)	At infinity (parallel rays focus on retina)

Classification of Refractive Errors

Type	Defect	Correction
Myopia	Image in FRONT of retina	Concave (minus) lens
Hypermetropia	Image BEHIND retina	Convex (plus) lens
Astigmatism	Two focal lines (no point focus)	Cylindrical lens
Presbyopia	Loss of accommodation (age >40)	Convex reading add (+1 to +3 D)

Presbyopia -- Age-wise Addition (High Yield!)

Age	Add Power	Near Point	Remark
40 years	+1.0 D	25 cm	Earliest symptom: difficulty reading
45 years	+1.5 D	33 cm	Most patients present at this age
50 years	+2.0 D	50 cm	Needs reading glasses for sure
55 years	+2.5 D	1 m	Near vision significantly affected
60 years	+3.0 D	Infinity	Maximum add needed (accommodation = 0)

★ **Remember: CORNEA is King!**
Cornea contributes 2/3 of refractive power. That's why LASIK reshapes the CORNEA, not the lens!

Myopia vs Hypermetropia comparison table = asked in EVERY university exam!

Myopia vs Hypermetropia -- The Comparison

Feature	Myopia (Short sight)	Hypermetropia (Long sight)
Image falls	In FRONT of retina	BEHIND the retina
Eye is	Too long / too powerful	Too short / too weak
Far point	Finite (nearer than infinity)	Behind the eye (virtual)
Lens for correction	Concave (diverging / minus)	Convex (converging / plus)
Commonest cause	Increased axial length	Decreased axial length
Vision affected	Distant vision blurred	Near vision affected first
Accommodation	Not needed for near	Constantly used (strain)
AC depth	Deep anterior chamber	Shallow anterior chamber
Angle closure risk	Low	HIGH (predisposed!)
Fundus changes	Myopic crescent, staphyloma	Usually normal

Types of Myopia

Type	Details
Simple myopia	Most common, <6 D, benign, axial type, stops by 20 yrs
Pathological myopia	>6 D, progressive, begins age 5-10, degenerative changes
Curvature myopia	Increased corneal/lenticular curvature (keratoconus)
Index myopia	Increased refractive index of lens (nuclear sclerosis, DM)
Positional myopia	Anterior displacement of lens

Astigmatism -- Types + Classification

Type	Description
With-the-rule (WTR)	Vertical meridian more curved (common in young)
Against-the-rule (ATR)	Horizontal meridian more curved (common in elderly)
Oblique	Principal meridians not at 90/180 degrees
Simple myopic	One focus on retina, other in front
Simple hypermetropic	One focus on retina, other behind
Compound myopic	Both foci in front of retina
Compound hypermetropic	Both foci behind retina
Mixed	One focus in front, one behind retina

WTR vs ATR Mnemonic
WTR = Young = Vertical steep (think: standing tall when young). ATR = Old = Horizontal (lying down!)

Correction Methods + Refractive Surgery

LASIK vs PRK vs SMILE
is a HOT exam topic now!
Know the differences!

Clinical Testing of Refraction

Test	Details
Retinoscopy	OBJECTIVE test. Uses retinoscope at 1m. Gold standard for children
Subjective refraction	Patient's response with trial lenses. Duochrome, JCC tests
Autorefractometer	Automated objective refraction. Screening tool
Keratometry	Measures corneal curvature. Essential pre-LASIK & IOL calc
A-scan biometry	Measures axial length. For IOL power calculation
Cycloplegic refraction	After atropine/cyclopentolate. MUST in children (<15 yrs)

Correction Methods -- Spectacles + Contact Lenses

Method	Advantages	Disadvantages
Spectacles	Safest, cheapest, easy	Cosmetic issues, prismatic effect
Soft contact lens	Cosmetic, good for sports	Infections, dry eye, corneal ulcer
RGP contact lens	Best for astigmatism, keratoconus	Discomfort initially, expensive
Orthokeratology	Overnight wear, reshapes cornea	Temporary effect, infection risk

Refractive Surgery -- The Comparison Table

Feature	LASIK	PRK/LASEK	SMILE
Flap	YES (microkeratome/FS)	NO flap	NO flap (lenticule)
Laser	Excimer (stromal bed)	Excimer (surface)	Femtosecond only
Pain post-op	Minimal	Moderate (2-3 days)	Minimal
Recovery	Fast (1-2 days)	Slower (5-7 days)	Fast (2-3 days)
Corneal strength	Reduced (flap risk)	Better preserved	Best preserved
Dry eye	More common	Less	Least
Range (myopia)	Up to -10 D	Up to -6 D	Up to -10 D
Best for	Most common procedure	Thin cornea, contact sport	High myopia, dry eye

★ LASIK Eligibility Criteria (Exam Favourite!)

- Age >18 years | Stable refraction for >1 year | Corneal thickness >500 microns
- No keratoconus/corneal ectasia | No active eye disease | Myopia up to -10 D
- Pupil size considered (>7mm = more glare) | Not pregnant/lactating

Complications of high myopia = LAQ favourite! Draw the diagram!

Complications of Pathological/Degenerative Myopia

ANTERIOR SEGMENT

- Deep anterior chamber
- Prominent eyes (proptosis-like)
- Vitreous degeneration & floaters
- Vitreous detachment (PVD)
- Lattice degeneration of retina

POSTERIOR SEGMENT

- Posterior staphyloma (MC complication)
- Myopic crescent (temporal)
- Foster-Fuchs spot (hemorrhage at macula)
- Retinal detachment (MOST SERIOUS)
- Choroidal neovascularization (CNVM)

Types of Hypermetropia (Donders' Classification)

Type	Description
Latent	Corrected by tone of ciliary muscle. Revealed only by cycloplegia
Manifest	Total hypermetropia minus latent = what patient uses
-- Facultative	Part of manifest corrected by accommodation (young patients)
-- Absolute	Cannot be overcome by accommodation. Always symptomatic
Total	Latent + Manifest. Found on cycloplegic refraction

★ **Retinoscopy -- The Light Reflex**

WITH movement = Hypermetropia (use + lens till neutral). AGAINST movement = Myopia (use - lens).

At 1 metre working distance: subtract 1 D from final reading. Neutral = no movement = endpoint!

Accommodation -- Quick Revision

Condition	Key Point
Spasm of accommodation	Pseudo-myopia. Common in students. Treat: cycloplegics
Paralysis of accommodation	Cycloplegia. Cause: III nerve palsy, drugs (atropine)
Insufficiency of accommodation	Premature presbyopia. Asthenopia in young patients
Accommodative esotropia	Hypermetropia -> excess accommodation -> convergence -> squint

Exam Strategy + High Yield

Refraction errors = guaranteed 15+ marks! You've GOT this Sammy!

Most Repeated LAQs (10-12 marks)

1. Classify refractive errors. Describe myopia -- types, clinical features, complications, Rx
2. Define astigmatism. Classify and discuss its management
3. Describe pathological myopia -- features, complications & management
4. Compare myopia vs hypermetropia (table format!)
5. Discuss refractive surgery -- LASIK, PRK. Indications & contraindications

Most Repeated SAQs (5-6 marks)

1. Types of astigmatism with correction
2. Retinoscopy -- principle & procedure
3. Presbyopia -- definition, age-wise addition table
4. Complications of contact lens use
5. Donders' classification of hypermetropia
6. LASIK -- indications, contraindications, complications

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
#2 cause of visual impairment in India	Refractive errors (19.7%)
Most common refractive error	Myopia
Most common cause of myopia	Increased axial length
Pathognomonic of degenerative myopia	Posterior staphyloma
Foster-Fuchs spot seen in	Degenerative myopia (macular hemorrhage)
Spasm of accommodation mimics	Myopia (pseudo-myopia)
Commonest type of astigmatism in youth	With-the-rule (WTR)
Retinoscopy: WITH movement means	Hypermetropia (use + lens)
Accommodative esotropia caused by	Uncorrected hypermetropia
Best objective test in children	Retinoscopy under cycloplegia
Keratoconus corrected best by	RGP contact lens
Night myopia due to	Spherical aberration + mydriasis in dark

High Yield Checklist

MUST KNOW

- Myopia vs Hypermetropia table
- Types of astigmatism
- Presbyopia age-wise add
- Retinoscopy principle
- LASIK criteria

HIGH YIELD

- Pathological myopia complications
- Donders' classification
- Accommodative esotropia
- Contact lens complications
- WTR vs ATR astigmatism

BONUS

- SMILE vs LASIK vs PRK
- SRK formula (IOL calc)
- Keratoconus management
- Night myopia mechanism
- Pseudo-myopia (spasm)

♥ Clear vision is a gift, Sammy! You're learning to give it to millions! ♥